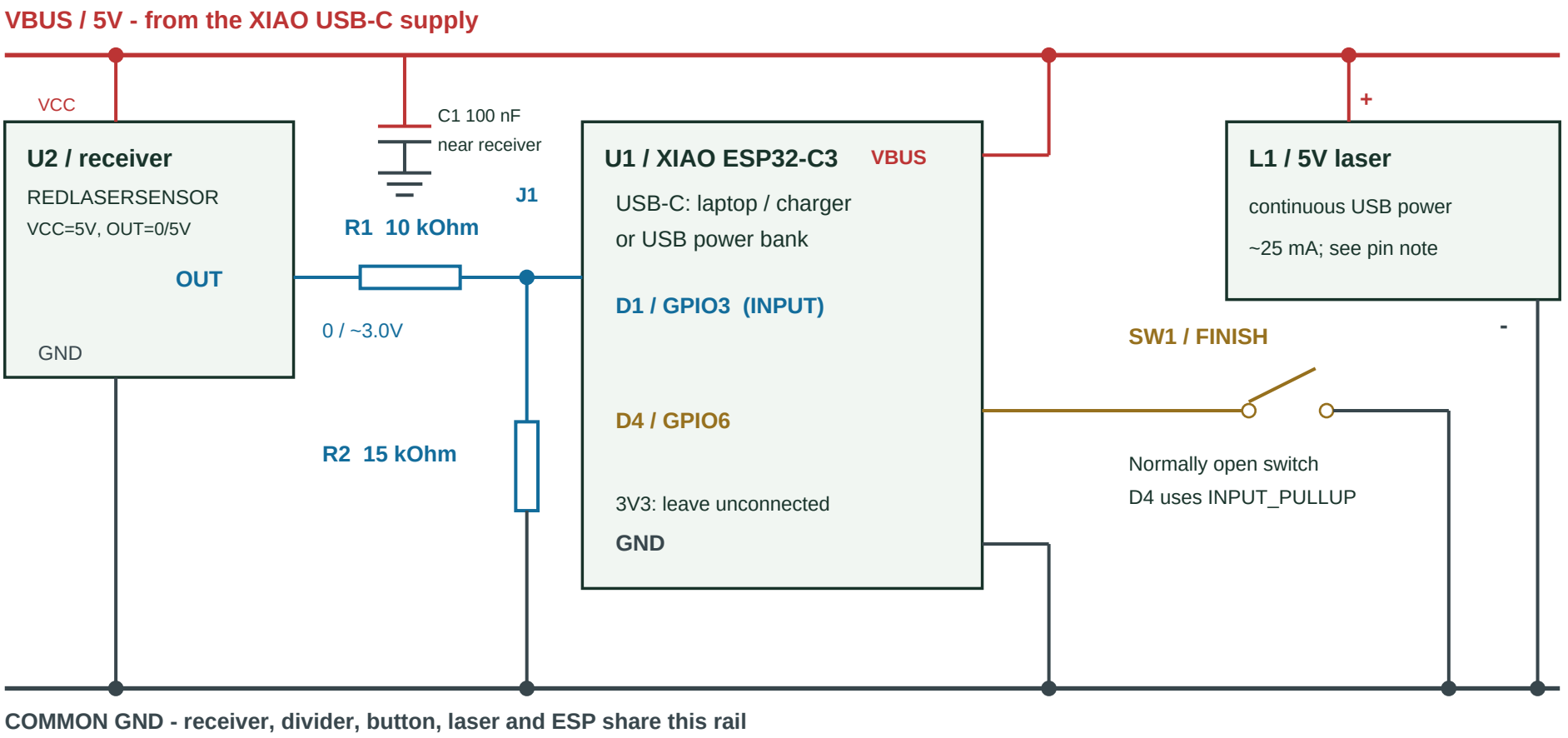
One beam. One finish button.

Confirmed receiver: TinyTronics REDLASERSENSOR / 001061. Board: Seeed XIAO ESP32-C3.

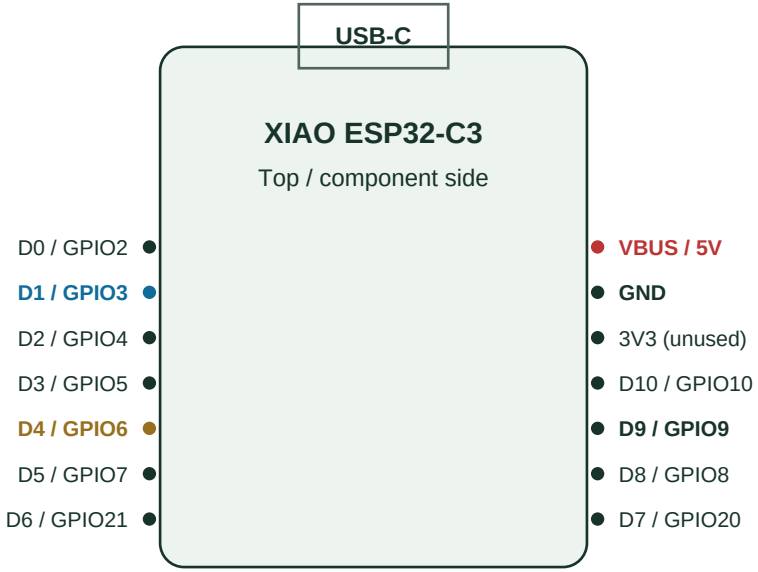


5V power is allowed on VBUS; 5V data must never go directly to D1. With a true 5V OUT, J1 is 3.0V. Measure the loaded HIGH: it must be 2.5-3.3V; LOW must be below 0.8V. If the receiver output sags, see the 100k/150k alternative in docs/LASER_WIRING.md.

Emitter identification: On TinyTronics **LSRMOD / 001339**, S is positive, - is negative and the middle pin is unused. On the two-wire **LSRDRD**, use its positive and negative wires. Connect the complete rated 5V module to VBUS/GND; no GPIO power connection.

Pin guide and first power-up

Read the printed labels. The receiver and laser emitter have different pin functions.



Parts: 1 x XIAO C3 + antenna; 1 x receiver; 1 x 5V emitter; 10k and 15k resistors; normally-open pushbutton; jumpers; USB-C cable. Recommended: 100nF ceramic across receiver VCC/GND.

4-leg button: two pins on each electrical side are already joined. D4 and GND must go to opposite electrical pairs. Pressing must connect them; releasing must open them.

Beam path: laser -> mirror(s) -> receiver. Keep the path below eye level and terminate it on a non-reflective backstop. Disconnect USB before moving wires.

1 / Check voltages first

Leave D1 disconnected. USB powers VBUS. Measure the divider junction with the beam aligned and blocked. HIGH must be 2.5-3.3V, LOW below 0.8V. If HIGH is too low, try 100k/150k, measure again, or use a proper level buffer.

2 / Match the firmware

Connect J1 to D1 after unplugging USB. LaserConfig.h uses D1=GPIO3, INPUT and blocked LOW. If your measured blocked state is HIGH, change brokenLow to false. The finish button is D4=GPIO6, active LOW.

3 / Align in the test page

Home -> Test your module -> Laser -> Connect test inputs -> Start alignment mode. The emitter stays powered continuously from VBUS. Watch detected/missing and input_raw while adjusting mirrors. State updates arrive about every 2 seconds.

4 / Check the heist

Leave alignment mode. Break/clear/break the beam: each fresh break should count once. Press finish once. Use one crew controller for a real round: each cut costs 5 seconds; finish completes Laser without resetting the shared clock.

Source checks: TinyTronics REDLASERSENSOR, LSRMOD and LSRDRD product pages; Seeed XIAO C3 pin map; Espressif ESP32-C3 datasheet section 5.4. Full links and upload commands: **docs/LASER_WIRING.md**. Wiring/polarity and physical operation still require a real-board check.